Career Launcher MBA

QA Numbers-04

Divisibility; Remainders

Question 1

The number 3798125P369 is divisible by 7. What is the value of the digit P?

(1) 1

(2) 6

(3) 7

(4) 9

Question 2

A digit $n > 3$ is divisible by 3 but not divisible by 6. Which one of the following is divisible by 4?

(1) $2n$

(2) $3n$

(3) $2n + 4$

(4) $3n + 1$

Question 3

If N = 192837465564738291, then find the remainder when (N + 1)(N + 2)… (N + 7) is divided by 11.

(1) 7

(2) 5

(3) 2

(4) 9

Question 4

The sum of the digits of a number A is 13. What is the remainder when A^4 is divided by 3?

(1) 1

(2) 2

(3) 0

(4) Either (1) or (2)

Question 5

A certain number, when divided by 899, leaves a remainder 64. Find the remainder when the same number is divided by 29.

(1) 6

(2) 5

(3) 1

(4) Cannot be determined

Question 6

Two numbers, x and y, are such that when divided by 6, they leave remainders 4 and 5 respectively. Find the remainder when $(x^2 + y^2)$ is divided by 6.

(1) 2

(2) 3

(3) 4

(4) 5

Question 7

S is the set of positive integers such that when divided by 2, 3, 4, 5, 6 leaves the remainders 1, 2, 3, 4, 5 respectively. How many integers between 0 and 200 belong to set S?

(1) 0

(2) 1

(3) 2

(4) 3

Question 8

When a number successively divided by 3, 4 and 7, the remainders obtained are 2, 1 and 4 respectively. What will be the remainder if 42 divides the same number?

(1) 11

(2) 13

(3) 39

(4) 41

Question 9

If 67516 and 68488 are divided by a three-digit number, they leave the same remainder. How many such three digits numbers are possible?

Question 10

Find the smallest three-digit number which when divided by 7 and 8, leaves remainder 2 and 4.

Question 11

What is the remainder when $(37)^{47}$ is divided by 19?

(1) 18

(2) 17

(3) 1

(4) 0

Question 12

What is the remainder when 4^{96} is divided by 6?

(1) 0

(2) 2

(3) 3

(4) 4

Question 13

Find the remainder when 5^{4k}, where k is a natural number, is divided by 6.

Question 14

Find the remainder when $(1^3 + 3^3 + 5^3 + ... + 23^3)$ is divided by 4.

(1) 0

(2) 1

(3) 3

(4) 2

Question 15

What is the remainder when $(15^{23} + 23^{23})$ is divided by 19?

(1) 4

(2) 15

(3) 0

(4) 18

Question 16

What is the remainder when $n! + (n! + 1) + (n! - 2) + (n! + 3) \ldots\ldots + (n! - 2006)$ is divided by 1003 for $n = 1003$?

(1) 1

(2) 0

(3) 2006

(4) None of these

Question 17

$7^{6n} – 6^{6n}$, where n is an integer greater than 0, is not divisible by

(1) 13

(2) 81

(3) 127

(4) 559

Question 18

$6^{6n} – 5^{6n}$, where ‘n’ is an integer greater than 0, is divisible by

(1) 11

(2) 91

(3) 341

(4) All (1), (2) and (3)

Question 19

$N = (5^p + 1)(5^q + 1)(5^r + 1)$ and $M = (5^p - 1)(5^q + 1)(5^r + 1)$, where p, q and r are consecutive natural numbers. What is the remainder when $(N \times M)$ is divided by 6?

(1) 0

(2) 1

(3) 5

(4) 4

Question 20

P is a natural number and $Q = 2^P$. What is the remainder when Q! is divided by 2^Q?

(1) $2^Q - 1$

(2) 2^{Q-1}

(3) $2^{Q-1} - 1$

(4) None Of these

Thanks!